ODAG Report

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Overview

This analysis seeks to evaluate the relationship between an African country's official development assistance gross (ODAG) and their democracy score. ODAG measures the aid received by an African country from external countries and private organizations. (Note: In this report, we refer to ODAG received from other countries as ODAG and ODAG received from entities that are not countries are ODAGprivate.) We used democracy scores from Freedom House (TOTLIB1), the POLITY IV database (POLITY2), and the Liberal Democracy Index from the V-Dem Institute (V2x_Libdem). (Note: Here we only analyze POLITY2, whereas in previous reports we have analyzed both POLITY and POLITY2.)

This main analyses were split into a reduced and full model. The reduced model only considers democracy as a fixed effect, while including random intercepts for the African country and year. These random effects account for structural correlation between observations within the data structure.

The full model includes all of the parts of the reduced model while adding the African country's population, gross national income (GNI), total imports, total exports, colonial Past, region, and a cold war indicator variable. We applied a log transformation to GNI, population, imports, and exports due to their right skew.

ODAG (Countries)

In the reduced models (Tables 1-3), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with larger levels of total ODAG.

In the full models (Table 4-6), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with high total ODAG. In addition, we see that total population, GNI, total imports, and the cold war indicator are all significant. Population and total imports were positively correlated with total ODAG

whereas larger GNI was related to lower levels of ODAG. Finally, the cold war (pre-1992) is associated with lower total levels of ODAG.

ODAG (Private)

In the reduced models (Tables 7-9), once again all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with larger levels of total ODAG.

In the full models (Table 10-12), all V2x_Libdem, POLITY2, and TOTLIB1 coefficients are again significant at $\alpha = 0.05$. In all cases, higher democracy scores were associated with high total ODAG. As before, we see that total population, GNI, total imports, and the cold war indicator are all significant. However, unlike the previous full models, total exports in also significant here. Population and total imports were positively correlated with total ODAG whereas larger GNI and total exports was related to lower levels of ODAG. Finally, the cold war indicator (pre-1992) is associated with lower total levels of ODAG.

Models

Reduced Model

The reduced model is described below, where the response Y_{it} is ODAG for African country i in year t. We applied a log transformation to ODAG for better model fit. The predictor X_{1it} represents the democracy score of African country i in year t.

In lagged models, the year t changes units depending on the lag. For instance, a one-year lag would produce the model equation $\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the external country's democracy score in the prior year (year t-1).

$$Y_{it} \sim \text{N}(\mu_{it}, \sigma^2)$$

$$\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$$

$$\gamma_{0i} \sim \text{N}(0, \sigma^2_{\text{country}})$$

$$\gamma_{0t} \sim \text{N}(0, \sigma^2_{\text{year}})$$

Full Model

Our full model equations are given below. They follow the same setup as the reduced model with the addition of the aforementioned transformed country characteristics. In lagged models, we only lagged the country's democracy score. For instance, a one-year lag would produce the model equation $\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1i(t-1)} + ... + \gamma_{0i} + \gamma_{0t} + \epsilon_{it}$, where $X_{1i(t-1)}$ is the country's democracy score in the prior year (year t-1).

Additional predictors in this model include the African country's log-transformed GNI, population, and total ODAG. Quadratic terms were included for the log-transformed GNI, population, and total ODAG. The log-transformed population variable added a cubic term to capture further nonlinear tendencies.

$$Y_{it} \sim \text{N}(\mu_{it}, \sigma^2)$$

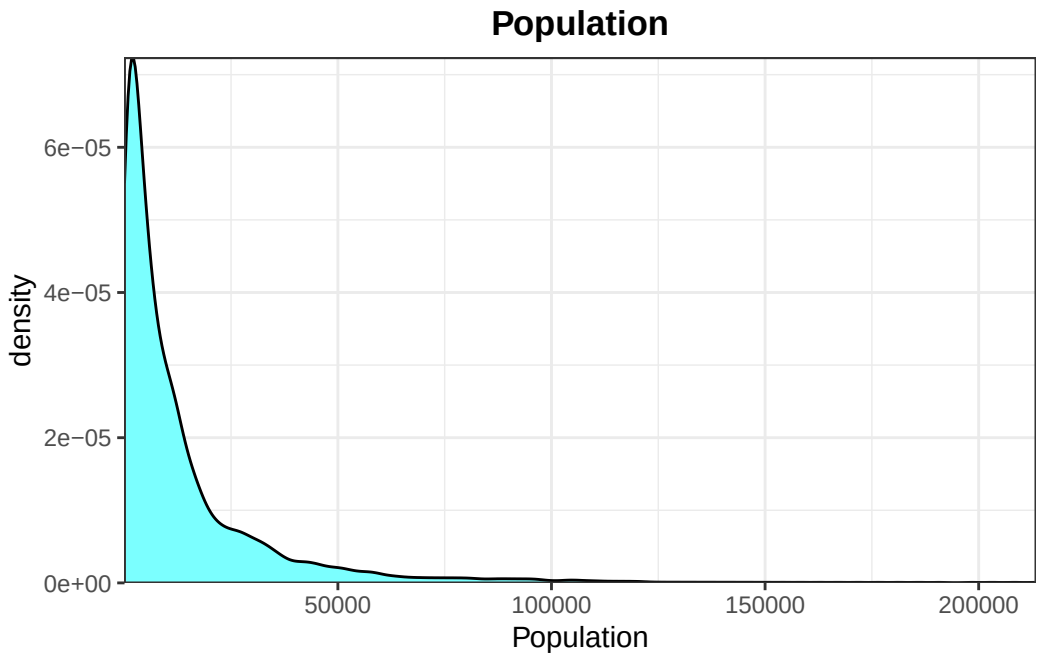
$$\log(\text{ODAG}_{it}) = \beta_0 + \beta_1 X_{1ik} + ... + \gamma_{0i} + \gamma_{tk} + \epsilon_{it}$$

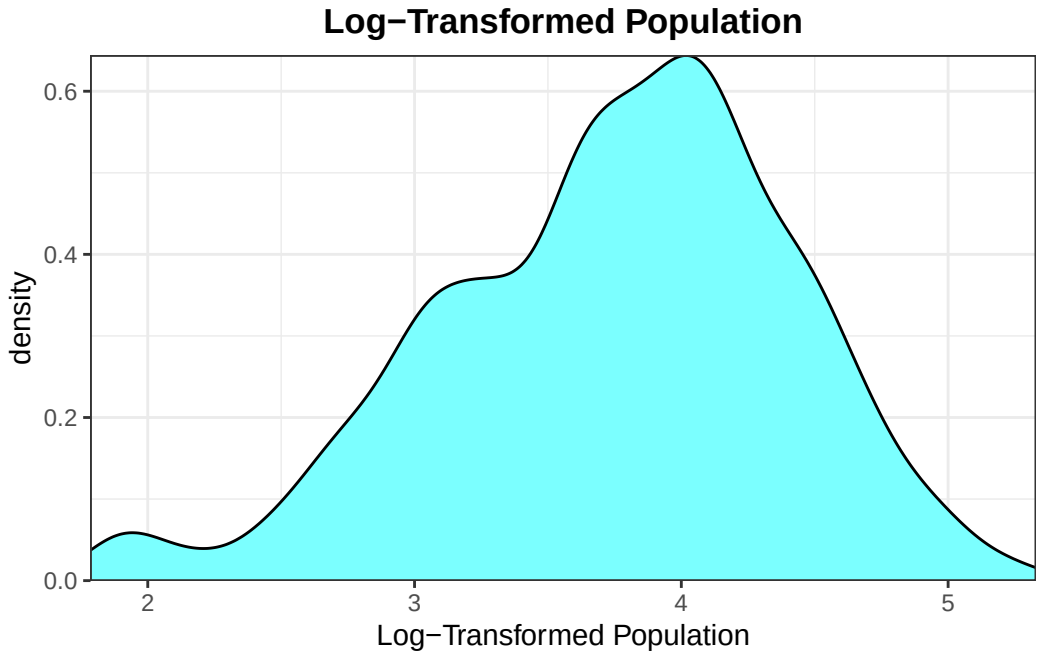
$$\gamma_{0i} \sim \text{N}(0, \sigma^2_{\text{country}})$$

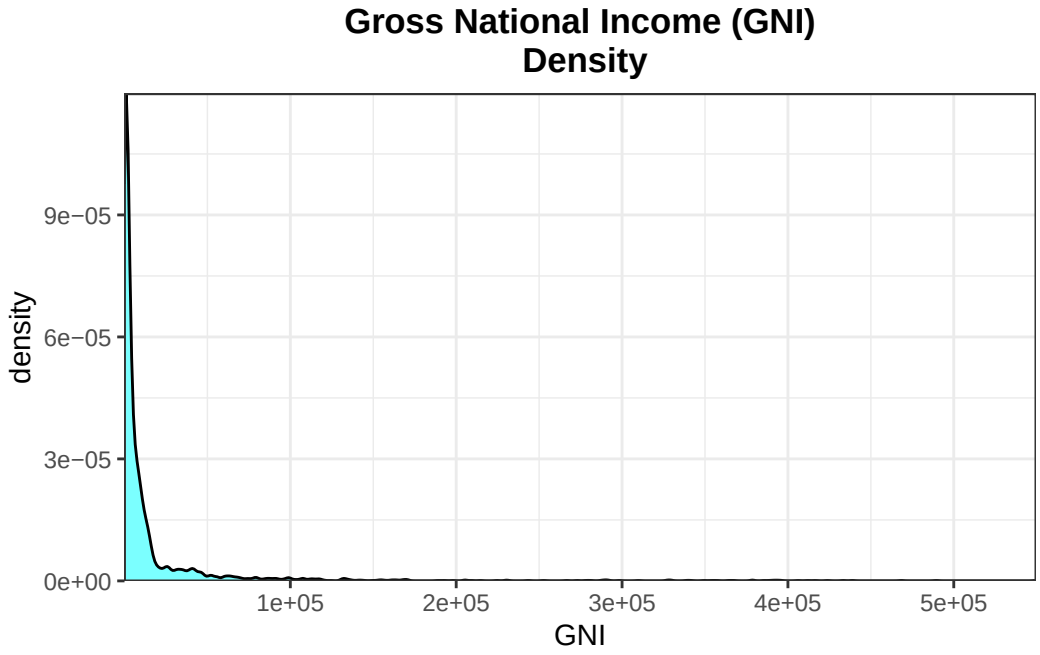
$$\gamma_{0t} \sim \text{N}(0, \sigma^2_{\text{year}})$$

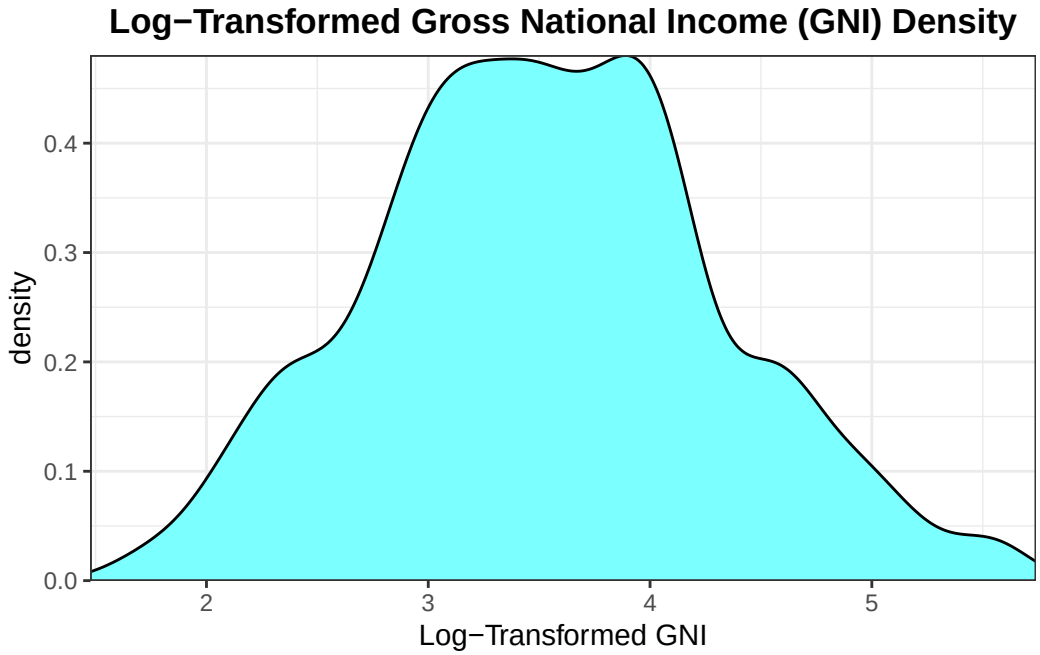
Data Visualizations and Results

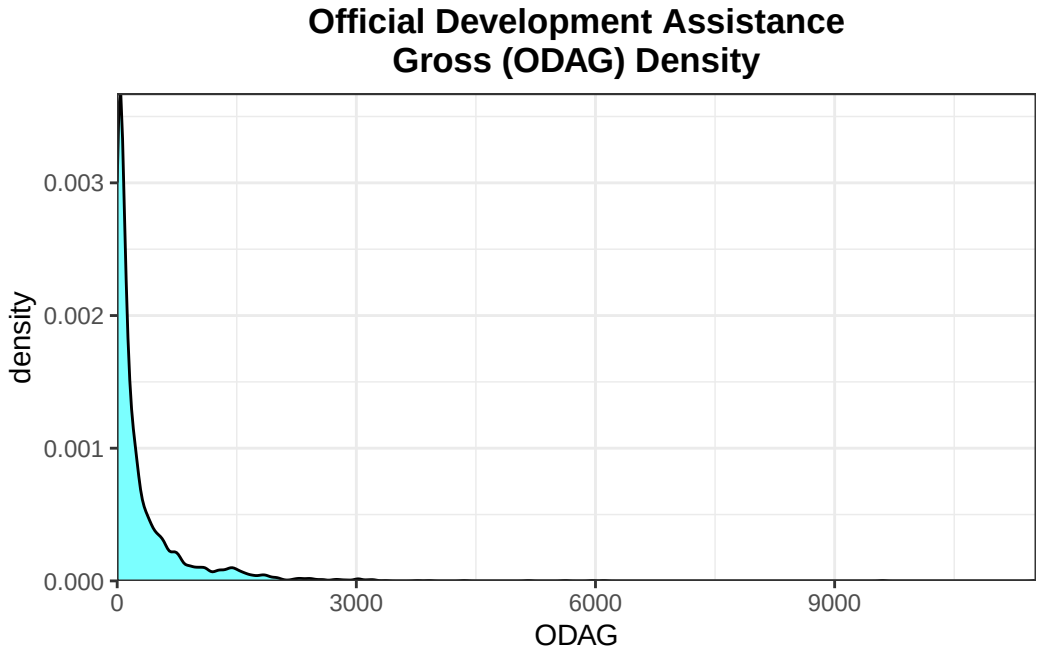
The distributions of the full model predictor variables and their log transformations are plotted below. The log-transformed variables all display a noticeably reduced right-skew, improving their performance and interpretation in the full models.

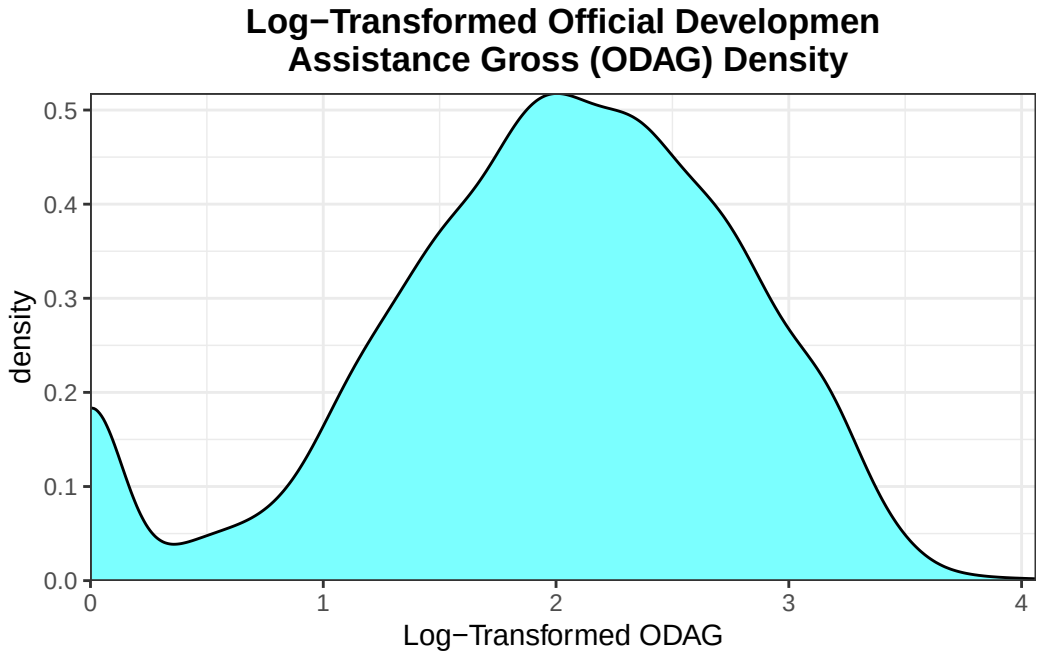

Population
6e−05
4e−05
2e−05
0e+00
density
50000
100000
150000
200000
Population

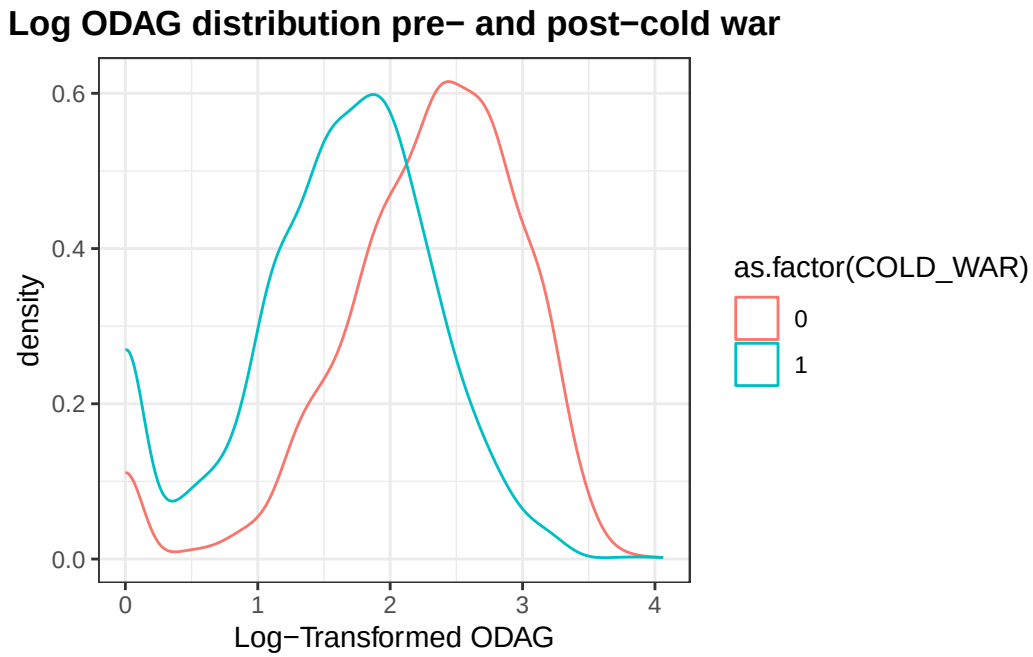

Log−Transformed Population
0.6
0.4
0.2
0.0
density
2
3
4
5
Log−Transformed Population

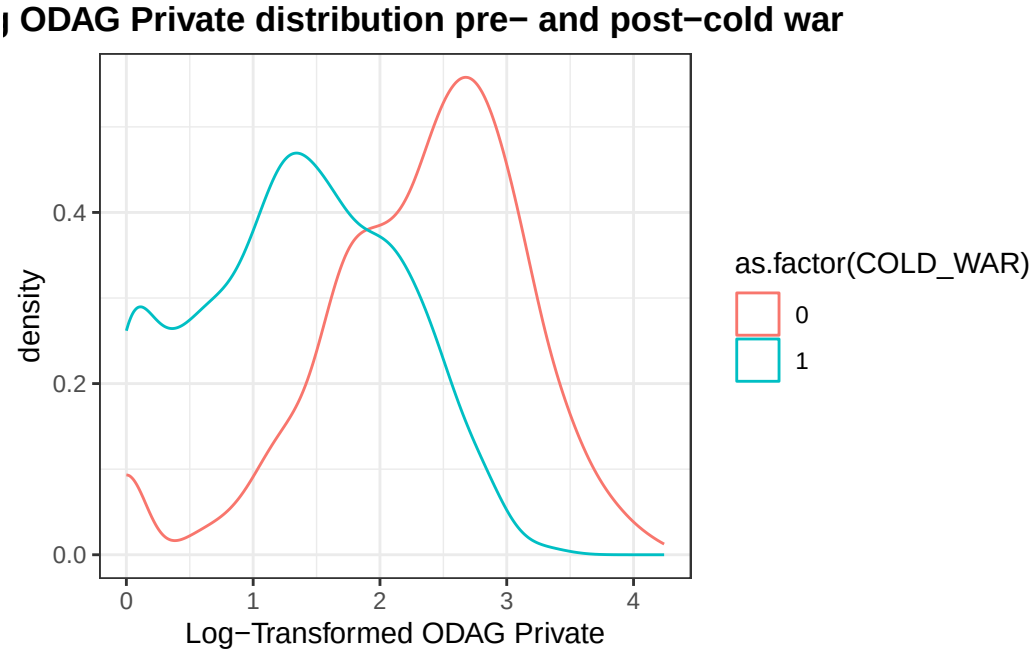

Gross National Income (GNI)
Density
density
9e−05
6e−05
3e−05
0e+00
1e+05
2e+05
3e+05
4e+05
5e+05
GNI


Log−Transformed Gross National Income (GNI) Density
density
0.4
0.3
0.2
0.1
0.0
2
3
4
5
Log−Transformed GNI


Official Development Assistance
Gross (ODAG) Density
density
0.003
0.002
0.001
0.000
0
3000
6000
9000
ODAG


Log−Transformed Official Developmen
Assistance Gross (ODAG) Density
density
0.5
0.4
0.3
0.2
0.1
0.0
0
1
2
3
4
Log−Transformed ODAG





Summaries for the reduced and full models for ODAG are shown with the regression coefficient estimates in the column labeled "Est." starred by significance as denoted in the legend below. each table. Next to each coefficient estimate is their respective standard error.

Table 1: **ODAG: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.492***	0.115	1.499***	0.113	1.527***	0.111	1.559***	0.108
V2x_Libdem	0.889***	0.069						
V2x_Libdem_lag1			0.903***	0.068				
V2x_Libdem_lag2					0.932***	0.068		
V2x_Libdem_lag3							0.959***	0.069
SD (Intercept YEAR)	0.791		0.781		0.749		0.710	
SD (Intercept CCODE)	0.506		0.500		0.504		0.507	
SD (Observations)	0.336		0.333		0.328		0.324	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

ODAG Results

Below are the summaries for the full models for ODAG from external countries. As with the reduced model summaries, the $\widehat{\beta}$ coefficients are starred by significance as denoted in each table, with their respective standard error below.

ODAG Private Results

Below are the summaries for the full models for ODAG from private organizations As with the reduced model summaries, the $\widehat{\beta}$ coefficients are starred by significance as denoted in each table, with their respective standard error below.

Table 2: **ODAG: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.657***	0.114	1.700***	0.112	1.742***	0.109	1.787***	0.106
POLITY2	0.008***	0.002						
POLITY2_lag1			0.007***	0.002				
POLITY2_lag2					0.006***	0.002		
POLITY2_lag3							0.005**	0.002
SD (Intercept YEAR)	0.812		0.785		0.756		0.721	
SD (Intercept CCODE)	0.434		0.438		0.441		0.442	
SD (Observations)	0.334		0.326		0.322		0.318	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: **ODAG: Reduced Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	2.453***	0.096	2.474***	0.096	2.468***	0.096	2.463***	0.096
TOTLIB1	−0.034***	0.003						
TOTLIB1_lag1			−0.035***	0.003				
TOTLIB1_lag2					−0.033***	0.003		
TOTLIB1_lag3							−0.031***	0.003
SD (Intercept YEAR)	0.424		0.415		0.408		0.403	
SD (Intercept CCODE)	0.516		0.518		0.517		0.517	
SD (Observations)	0.317		0.315		0.316		0.317	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: **ODAG: Full Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−0.186	0.297	−0.235	0.301	−0.265	0.301	−0.228	0.295
V2x_Libdem	0.863***	0.065						
V2x_Libdem_lag1			0.888***	0.064				
V2x_Libdem_lag2					0.899***	0.064		
V2x_Libdem_lag3							0.881***	0.064
logPOP	0.703***	0.062	0.702***	0.063	0.693***	0.063	0.690***	0.063
logGNI	−0.143***	0.032	−0.140***	0.032	−0.133***	0.031	−0.142***	0.032
logTRDIMtot	0.131***	0.020	0.142***	0.020	0.154***	0.020	0.146***	0.021
logTRDEXtot	−0.041*	0.019	−0.046*	0.019	−0.049*	0.019	−0.035+	0.020
COLD_WAR	−0.396***	0.083	−0.379***	0.081	−0.358***	0.078	−0.351***	0.073
CW_Socialist	−0.038	0.035	−0.017	0.035	−0.010	0.035	−0.011	0.035
CW_Marxist	0.003	0.037	0.000	0.036	−0.006	0.036	−0.015	0.036
COL_British	−0.104	0.105	−0.109	0.107	−0.116	0.108	−0.121	0.106
COL_Portuguese	0.060	0.146	0.061	0.149	0.056	0.150	0.055	0.147
COL_Spanish	−0.295	0.292	−0.296	0.298	−0.306	0.299	−0.327	0.293
COL_Italian	−0.169	0.215	−0.147	0.219	−0.138	0.220	−0.137	0.216
COL_Belgian	0.079	0.297	0.060	0.303	0.041	0.304	0.016	0.298
COL_German	0.000	0.135	−0.006	0.137	−0.015	0.138	−0.018	0.135
COL_US	0.147	0.280	0.149	0.286	0.144	0.287	0.122	0.281
COL_None	−0.162	0.297	−0.167	0.303	−0.163	0.304	−0.151	0.298
COL_Unique	0.321	0.302	0.316	0.307	0.315	0.308	0.318	0.302
REG_E_Africa	−0.039	0.159	−0.016	0.162	−0.001	0.162	−0.002	0.159
REG_S_Africa	−0.186	0.170	−0.165	0.173	−0.142	0.174	−0.138	0.170
REG_C_Africa	−0.167	0.175	−0.141	0.178	−0.115	0.179	−0.111	0.176
REG_W_Africa	−0.251+	0.150	−0.230	0.153	−0.208	0.154	−0.203	0.151
SD (Intercept YEAR)	0.303		0.294		0.278		0.260	
SD (Intercept CCODE)	0.264		0.270		0.271		0.266	
SD (Observations)	0.284		0.279		0.274		0.272	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 5: **ODAG: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.209	0.332	0.181	0.327	0.201	0.319	0.272	0.308
POLITY2	0.008***	0.002						
POLITY2_lag1			0.007***	0.002				
POLITY2_lag2					0.006***	0.002		
POLITY2_lag3							0.005**	0.002
logPOP	0.638***	0.073	0.638***	0.072	0.612***	0.071	0.598***	0.069
logGNI	−0.126***	0.034	−0.131***	0.034	−0.123***	0.033	−0.134***	0.033
logTRDIMtot	0.131***	0.022	0.145***	0.022	0.162***	0.022	0.154***	0.022
logTRDEXtot	−0.027	0.021	−0.032	0.021	−0.037+	0.021	−0.020	0.021
COLD_WAR	−0.489***	0.088	−0.478***	0.083	−0.457***	0.077	−0.452***	0.071
CW_Socialist	−0.036	0.037	−0.013	0.037	0.012	0.037	0.022	0.037
CW_Marxist	−0.016	0.039	−0.022	0.038	−0.026	0.038	−0.034	0.037
COL_British	−0.086	0.108	−0.089	0.107	−0.087	0.106	−0.079	0.103
COL_Portuguese	0.099	0.161	0.089	0.160	0.073	0.157	0.068	0.153
COL_Spanish	−0.386	0.294	−0.410	0.293	−0.447	0.288	−0.488+	0.280
COL_Italian	−0.315	0.215	−0.291	0.214	−0.280	0.211	−0.276	0.205
COL_Belgian	0.066	0.297	0.046	0.295	0.042	0.291	0.031	0.282
COL_German	0.043	0.134	0.035	0.133	0.029	0.131	0.031	0.127
COL_US	0.153	0.278	0.144	0.277	0.130	0.273	0.098	0.264
COL_None	−0.179	0.297	−0.180	0.295	−0.156	0.291	−0.123	0.282
COL_Unique	0.250	0.309	0.257	0.303	0.321	0.299	0.350	0.288
REG_E_Africa	−0.083	0.159	−0.056	0.158	−0.040	0.155	−0.040	0.151
REG_S_Africa	−0.189	0.174	−0.161	0.173	−0.141	0.170	−0.138	0.165
REG_C_Africa	−0.207	0.177	−0.180	0.176	−0.156	0.174	−0.153	0.168
REG_W_Africa	−0.234	0.152	−0.209	0.151	−0.188	0.149	−0.182	0.144
SD (Intercept YEAR)	0.310		0.293		0.269		0.246	
SD (Intercept CCODE)	0.261		0.260		0.257		0.249	
SD (Observations)	0.294		0.289		0.285		0.282	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 6: **ODAG: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.328	0.290	0.281	0.293	0.291	0.289	0.301	0.284
TOTLIB1	−0.034***	0.003						
TOTLIB1_lag1			−0.036***	0.003				
TOTLIB1_lag2					−0.034***	0.003		
TOTLIB1_lag3							−0.032***	0.003
logPOP	0.728***	0.064	0.734***	0.065	0.720***	0.065	0.706***	0.064
logGNI	−0.211***	0.034	−0.206***	0.034	−0.199***	0.034	−0.193***	0.035
logTRDIMtot	0.134***	0.021	0.159***	0.022	0.157***	0.023	0.149***	0.023
logTRDEXtot	0.009	0.020	−0.014	0.021	−0.012	0.022	−0.002	0.022
COLD_WAR	−0.218***	0.053	−0.187***	0.051	−0.167***	0.048	−0.151**	0.046
CW_Socialist	0.020	0.043	0.033	0.043	0.025	0.044	0.019	0.044
CW_Marxist	0.045	0.038	0.054	0.038	0.046	0.039	0.035	0.039
COL_British	−0.097	0.105	−0.098	0.106	−0.096	0.105	−0.094	0.103
COL_Portuguese	0.014	0.146	0.015	0.148	0.020	0.146	0.027	0.143
COL_Spanish	−0.356	0.291	−0.317	0.296	−0.323	0.292	−0.356	0.285
COL_Italian	−0.153	0.214	−0.135	0.218	−0.145	0.215	−0.155	0.210
COL_Belgian	−0.065	0.296	−0.065	0.301	−0.061	0.297	−0.062	0.290
COL_German	0.015	0.134	0.013	0.137	0.017	0.135	0.022	0.132
COL_US	−0.019	0.280	0.006	0.285	0.015	0.281	0.019	0.274
COL_None	−0.196	0.296	−0.208	0.301	−0.194	0.297	−0.177	0.290
COL_Unique	0.457	0.300	0.462	0.304	0.466	0.300	0.469	0.294
REG_E_Africa	0.007	0.158	0.014	0.160	0.011	0.158	0.010	0.155
REG_S_Africa	−0.118	0.169	−0.108	0.172	−0.107	0.170	−0.105	0.166
REG_C_Africa	−0.096	0.175	−0.083	0.177	−0.087	0.175	−0.089	0.171
REG_W_Africa	−0.182	0.150	−0.173	0.152	−0.167	0.150	−0.159	0.147
SD (Intercept YEAR)	0.161		0.151		0.139		0.130	
SD (Intercept CCODE)	0.263		0.268		0.264		0.258	
SD (Observations)	0.272		0.270		0.272		0.272	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 7: **ODAG Private: Reduced Model Summary for V2X_LIBDEM**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.370***	0.123	1.389***	0.122	1.423***	0.122	1.451***	0.121
V2x_Libdem	1.064***	0.071						
V2x_Libdem_lag1			1.008***	0.072				
V2x_Libdem_lag2					0.962***	0.073		
V2x_Libdem_lag3							0.945***	0.074
SD (Intercept YEAR)	0.849		0.847		0.833		0.818	
SD (Intercept CCODE)	0.541		0.534		0.537		0.540	
SD (Observations)	0.348		0.350		0.349		0.349	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 8: **ODAG Private: Reduced Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	1.542***	0.120	1.586***	0.119	1.627***	0.119	1.663***	0.118
POLITY2	0.010***	0.002						
POLITY2_lag1			0.009***	0.002				
POLITY2_lag2					0.008***	0.002		
POLITY2_lag3							0.007***	0.002
SD (Intercept YEAR)	0.857		0.842		0.829		0.819	
SD (Intercept CCODE)	0.455		0.461		0.470		0.476	
SD (Observations)	0.351		0.349		0.347		0.346	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 9: **ODAG Private: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	2.458***	0.109	2.464***	0.108	2.446***	0.108	2.447***	0.108
TOTLIB1	−0.038***	0.003						
TOTLIB1_lag1			−0.036***	0.003				
TOTLIB1_lag2					−0.033***	0.003		
TOTLIB1_lag3							−0.031***	0.003
SD (Intercept YEAR)	0.506		0.489		0.479		0.471	
SD (Intercept CCODE)	0.564		0.566		0.568		0.569	
SD (Observations)	0.341		0.341		0.340		0.340	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 10: **ODAG Private: Full Model Summary for V2X_LIBDEM**

	Unlagged	One-Year Lag	Two-Year Lag	Three-Year Lag
	Est.	Est.	Est.	Est.
Intercept	−0.558+	−0.512	−0.488	−0.455
V2x_Libdem	1.113***			
V2x_Libdem_lag1		1.057***		
V2x_Libdem_lag2			0.989***	
V2x_Libdem_lag3				0.959***
logPOP	0.914***	0.920***	0.918***	0.923***
logGNI	−0.310***	−0.324***	−0.329***	−0.337***
logTRDIMtot	0.192***	0.192***	0.201***	0.201***
logTRDEXtot	−0.145***	−0.142***	−0.146***	−0.146***
COLD_WAR	−0.792***	−0.806***	−0.821***	−0.838***
CW_Socialist	−0.089*	−0.085*	−0.082*	−0.086*
CW_Marxist	−0.051	−0.044	−0.046	−0.048
COL_British	−0.185	−0.173	−0.161	−0.155
COL_Portuguese	−0.005	0.000	0.004	0.006
COL_Spanish	−0.254	−0.265	−0.275	−0.277
COL_Italian	−0.283	−0.277	−0.279	−0.279
COL_Belgian	−0.020	−0.029	−0.042	−0.041
COL_German	−0.058	−0.058	−0.056	−0.055
COL_US	−0.142	−0.142	−0.134	−0.130
COL_None	0.022	0.014	0.012	0.015
COL_Unique	−0.400	−0.410	−0.417	−0.421
REG_E_Africa	0.274	0.275	0.278	0.268
REG_S_Africa	0.051	0.044	0.045	0.040
REG_C_Africa	0.189	0.180	0.181	0.168
REG_W_Africa	0.188	0.185	0.191	0.185
SD (Intercept YEAR)	0.456	0.454	0.454	0.459
SD (Intercept CCODE)	0.288	0.284	0.281	0.279
SD (Observations)	0.305	0.306	0.306	0.307

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 11: **ODAG Private: Full Model Summary for POLITY2**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.073	0.325	0.042	0.321	−0.033	0.323	−0.047	0.321
POLITY2	0.006**	0.002						
POLITY2_lag1			0.005**	0.002				
POLITY2_lag2					0.005**	0.002		
POLITY2_lag3							0.003+	0.002
logPOP	0.868***	0.071	0.873***	0.070	0.875***	0.071	0.881***	0.070
logGNI	−0.365***	0.037	−0.363***	0.037	−0.350***	0.037	−0.353***	0.037
logTRDIMtot	0.218***	0.024	0.215***	0.024	0.224***	0.024	0.224***	0.025
logTRDEXtot	−0.145***	0.023	−0.140***	0.023	−0.146***	0.023	−0.146***	0.024
COLD_WAR	−0.935***	0.129	−0.940***	0.125	−0.940***	0.121	−0.957***	0.119
CW_Socialist	−0.104*	0.041	−0.097*	0.041	−0.081+	0.041	−0.078+	0.042
CW_Marxist	−0.085*	0.043	−0.082+	0.042	−0.084*	0.042	−0.087*	0.042
COL_British	−0.152	0.100	−0.142	0.099	−0.138	0.101	−0.128	0.101
COL_Portuguese	−0.044	0.148	−0.033	0.148	−0.029	0.150	−0.023	0.149
COL_Spanish	−0.405	0.272	−0.386	0.271	−0.395	0.275	−0.401	0.274
COL_Italian	−0.476*	0.200	−0.450*	0.199	−0.439*	0.202	−0.434*	0.202
COL_Belgian	0.011	0.274	−0.011	0.273	−0.040	0.277	−0.039	0.276
COL_German	0.002	0.124	−0.002	0.123	−0.010	0.125	−0.011	0.125
COL_US	−0.156	0.256	−0.159	0.255	−0.152	0.259	−0.148	0.258
COL_None	0.012	0.275	0.004	0.273	−0.008	0.277	−0.011	0.276
COL_Unique	−0.532+	0.292	−0.533+	0.287	−0.424	0.291	−0.412	0.287
REG_E_Africa	0.178	0.148	0.198	0.147	0.232	0.149	0.239	0.149
REG_S_Africa	0.059	0.161	0.059	0.160	0.074	0.163	0.076	0.162
REG_C_Africa	0.060	0.164	0.073	0.164	0.105	0.166	0.107	0.166
REG_W_Africa	0.164	0.141	0.178	0.140	0.209	0.142	0.218	0.142
SD (Intercept YEAR)	0.476		0.462		0.450		0.445	
SD (Intercept CCODE)	0.240		0.239		0.243		0.242	
SD (Observations)	0.320		0.320		0.320		0.320	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 12: **ODAG Private: Full Model Summary for TOTLIB1**

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	−0.177	0.311	−0.277	0.314	−0.344	0.316	−0.365	0.317
TOTLIB1	−0.039***	0.004						
TOTLIB1_lag1			−0.038***	0.004				
TOTLIB1_lag2					−0.034***	0.004		
TOTLIB1_lag3							−0.032***	0.004
logPOP	0.934***	0.069	0.934***	0.070	0.920***	0.071	0.909***	0.071
logGNI	−0.296***	0.038	−0.283***	0.039	−0.266***	0.039	−0.259***	0.040
logTRDIMtot	0.189***	0.024	0.215***	0.025	0.219***	0.025	0.214***	0.026
logTRDEXtot	−0.112***	0.023	−0.135***	0.024	−0.137***	0.024	−0.130***	0.025
COLD_WAR	−0.458***	0.068	−0.412***	0.062	−0.384***	0.058	−0.364***	0.056
CW_Socialist	−0.092+	0.048	−0.097*	0.049	−0.102*	0.049	−0.106*	0.050
CW_Marxist	0.034	0.043	0.028	0.043	0.011	0.044	−0.006	0.044
COL_British	−0.137	0.110	−0.133	0.112	−0.126	0.113	−0.121	0.113
COL_Portuguese	0.009	0.153	0.018	0.156	0.033	0.158	0.041	0.158
COL_Spanish	−0.227	0.306	−0.197	0.311	−0.209	0.315	−0.238	0.315
COL_Italian	−0.278	0.226	−0.262	0.229	−0.268	0.232	−0.270	0.233
COL_Belgian	−0.077	0.312	−0.083	0.317	−0.083	0.321	−0.081	0.321
COL_German	0.002	0.141	−0.005	0.144	−0.004	0.145	0.001	0.145
COL_US	−0.130	0.294	−0.099	0.299	−0.081	0.303	−0.080	0.303
COL_None	−0.045	0.311	−0.067	0.316	−0.067	0.320	−0.057	0.321
COL_Unique	−0.201	0.317	−0.213	0.322	−0.224	0.325	−0.226	0.326
REG_E_Africa	0.314+	0.167	0.344*	0.169	0.366*	0.171	0.377*	0.172
REG_S_Africa	0.103	0.178	0.126	0.181	0.142	0.183	0.153	0.183
REG_C_Africa	0.228	0.184	0.253	0.187	0.266	0.189	0.272	0.189
REG_W_Africa	0.280+	0.158	0.305+	0.161	0.330*	0.162	0.346*	0.163
SD (Intercept YEAR)	0.216		0.189		0.171		0.161	
SD (Intercept CCODE)	0.276		0.281		0.284		0.285	
SD (Observations)	0.307		0.307		0.307		0.307	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001